PROJECT TRICORDER

MK I · FIELD OPERATIONS MANUAL

On-device multimodal field instrument: inquiry, vision, image synthesis, neural speech in/out, offline translator (≥50 languages), audio scene analysis, avian species ID, flora/fauna classification, artificial horizon with G-meter and GPS speed, Celestrak/SGP4 satellite plot, major-star map, NOAA / flight / civic watch, append-only field log, and optional short-range tricorder-to-tricorder mesh.

Field	Value
Software version	0.6.4 (versionCode 39)
Application ID	com.projecttricorder.app
Platform	Android 8.0+ (API 26), arm64-v8a
Reference hardware	Samsung Galaxy Z Fold3 (Snapdragon 888, 12 GB)
Document date	5 September 2026
Classification	Unclassified — operator manual
Companion artifact	Tricorder-0.6.4-sky.apk
Supersedes	0.5.4 operations manual (4 September 2026)
Design	Jim Ames, Claude, and Grok · N2NHU Labs · Newburgh, New York 12550
Edition name	RENT-A-HAL Android Edition, implemented as Project Tricorder Mk I

NAVIGATION AND CERTIFICATION WARNING

This instrument is NOT certified by the FAA, NASA, EASA, ICAO, or any civil or military aviation or spaceflight authority. The artificial horizon, heading rose, G-meter, GPS ground-speed tape, accelerometer readout, barometer, satellite look-angles, and star map are uncertified consumer-sensor and public-catalog visualizations. They MUST NEVER be used as primary or backup navigation, attitude, approach, air-data, or celestial-navigation references for any aircraft, rotorcraft, UAV, or spacecraft. Do not use Project Tricorder to land, taxi, fly, dock, station-keep, or time a satellite pass for safety-of-flight. Treat the horizon and sky plot as field curiosities and demonstrations of on-device IMU / GNSS / SGP4 fusion only.

1. Purpose and scope

This manual is the operator and installer document for Project Tricorder Mk I on Android. It covers installation of the APK, the optional Termux inference stack, first-run pack fetch, every face control through software 0.6.4, theory of operations, capabilities, a competitive matrix, and troubleshooting.

The instrument is a field logger and multimodal companion computer. It is not a certified avionics, marine, or survey product. Imagine is a generative phrasebook for when there is no shared language and no clip-art binder — not a desktop art studio.

2. System overview

2.1 What the operator holds

The APK is the instrument face: WebView shell, on-device SpeechRecognizer, YAMNet, optional BirdNET and Coral flora/fauna TFLite packs, ML Kit translator, Nearby mesh, ContextWatch (NOAA / OpenSky / civic CAP +

USGS), GPS speed, IMU horizon, Celestrak TLE store, and the field book. Termux (F-Droid) hosts four local HTTP servers when the operator wants chat, vision, Imagine, and Kokoro.

2.2 Operating modes

Control	Function
I Inquiry	Gemma chat; Imagine keywords; COMPUTER easter egg; exact HELP opens the briefing
II Sensor	Camera + SmolVLM + flora/fauna on Hold
III Sentry	Motion then caption; Hold after each hit unless Continuous is on
IV Audio	YAMNet duty cycle, BirdNET on bird-like hits, watches if checked
T Translator	Offline speech-to-speech; SOURCE / TARGET wheels; AUTO swap
H Horizon	Green = tape on (idle + Audio). Red = off so the CRT can show the log
S Sat/Stars	Cycles satellites → stars → off. Polar live plot
L / E / V / P	Log review slider, officer note, video log, play current entry

2.3 What 0.6.x added over 0.5.4

Help / Credits / Warning / on-device Manual links. First-run “Initializing tricorder” with earth-link pack fetch (BirdNET, flora/fauna, en+es translator pair, Keplerian TLE groups) and Termux detect. Civic/public-safety watch via NWS IPAWS CAP + USGS (Citizen-style; not the Citizen Inc. product). Play-log for video (system player) and text (Kokoro). Horizon remapped for portrait hold. Live G-meter (load factor |a|/9.80665). GPS ground speed on the tape. SAT/STARS button with Celestrak + satellite.js SGP4 and a major-star alt-az map that follows compass heading. Audio-sense no longer speaks over a watch briefing.

3. Architecture

3.1 Process map

Kotlin MainActivity hosts a single WebView and a JavascriptInterface bridge. In-process: TFLite Task Audio (YAMNet), optional BirdNET and flora Interpreters, ML Kit Translation, SpeechRecognizer, MediaPlayer for Kokoro WAVs, LocationManager for speed and observer fix, Nearby Connections, ContextWatch pollers. Out-of-process (Termux): llama-server :8080 (Gemma) and :8081 (SmolVLM), sd-server :8188, Kokoro :5000.

3.2 Local ports

Port	Service	Role
8080	llama-server	Chat / inquiry
8081	llama-server + mmproj	Vision captions
8188	sd-server	Imagine (SD Turbo)
5000	kokoro-onnx	Speech out
8022	sshd (optional)	Desk SSH into Termux

3.3 Storage map

Path	Contents
files/logs/field.jsonl	Unified field book

files/logs/snaps/	JPEG frames for sensor/sentry/flora
files/logs/video/	Officer video logs
files/models/	BirdNET + Coral flora/fauna TFLite
files/tle/	Celestrak TLE packs (amateur, Iridium, …)
ML Kit cache	Downloaded translate language pairs

4. Theory of operations

4.1 Tuple routing

Inquiry text is matched against local tuples before Gemma. HELP and COMPUTER are intercepted on-device. Imagine keywords route to :8188. Everything else goes to :8080. Kokoro speaks the visible reply unless Continuous vision is on.

4.2 Inquiry path

Mic uses Android SpeechRecognizer. Text is shown, sent, then cleared. CHAT lamp is yellow while Gemma runs, VOICE yellow while Kokoro renders. Exact word HELP opens the Help sheet and Kokoro reads a short briefing that points at this manual.

4.3 Sensor and Sentry

Hold is a toggle. Red Hold freezes the frame and runs flora/fauna on that JPEG. Green resumes. Continuous machine vision writes to the CRT only and does not call Kokoro. After a Sentry hit the face drops into Hold so the stack is not flooded.

4.4 Audio path

Duty-cycle listen / sleep sliders. YAMNet classifies; BirdNET runs only when a bird-like label is already high. While Kokoro is playing a watch or inquiry line, Audio Sense holds so the loudspeaker is not classified as Speech. In Audio mode the LED tape keeps the detection line and does not crawl average AI service times.

4.5 Translator path

ML Kit on-device packs. Silence after speech triggers translate then device TTS (or Kokoro if wired). AUTO green swaps SOURCE and TARGET after each line. Reloading packs recovers a stuck recognizer.

4.6 Watch path (Audio + network)

NOAA NWS alerts, OpenSky states in a local box, and civic CAP events (AMBER, civil emergency, shelter, evac, fire, plus USGS M≥3.5 within 200 km) poll only while IV is armed, a watch box is on, and a network path exists. Hits log, speak, and push on Camp link. Waze has no public keyless feed and is not polled. Citizen Inc. has no public API; the civic box is NWS IPAWS + USGS labeled source=citizen.

4.7 Horizon, G-meter, and speed

The tape is drawn for a portrait hold (face toward the operator, top toward the sky). Device gravity is remapped so +Y is up the glass; screen-orientation remaps if the unit is rotated. Pitch and roll come from atan2 on that frame. Load factor G = |accelerationIncludingGravity| / 9.80665. Sitting still reads ~1.00 G. The right-hand tape is 0–3 G with a 1 G tick; the needle turns amber above ~1.6 G and red above ~2.4 G.

SPD is GPS ground speed from LocationManager (and a WebView geolocation fallback). It is not indicated airspeed and not IMU integration. No fix: SPD —. Below ~0.3 m/s the tape pins at 0.0. Units are m/s and km/h.

4.8 Satellites and stars

At earth-link init the APK fetches Celestrak GP/TLE groups (same public catalog Heavens-Above and Gpredict use): stations/ISS, amateur Part 97, Iridium NEXT, Globalstar, Orbcomm, GEO/Inmarsat-class, GPS operational, 100 visually brightest. The full Starlink catalog is not pulled by default (thousands of rows). satellite.js 4.1.4 (SGP4/SDP4) ships in the APK and propagates each enabled object to look-angles at the last GPS fix. The polar plot is zenith-center, horizon-rim; only elevation ≥ 0 is drawn. Group checkboxes gate which packs are propagated. Highest bird is labeled with range.

Second press of S switches to a major-star field (~40 catalog entries, Sirius through Hamal class). RA/Dec is converted to alt-az from observer lat/lon and UTC. The chart rotates with compass heading so turning the body moves the vault. Third press turns the plot off.

4.9 Camp link path

Nearby Connections, strategy P2P_STAR. Payload is JSON only. Media is not streamed. Both units must have Camp link on.

5. Capabilities and limits

Capability	Where	Network	Limit
Chat (Gemma 3 4B)	Termux :8080	None after weights	Seconds to tens of seconds
Vision (SmolVLM2)	Termux :8081	None after weights	Hold recommended
Imagine (SD Turbo)	Termux :8188	None after weights	256–512 px, 4–8 steps; tens of seconds to minutes
Kokoro TTS	Termux :5000	None after ONNX	Full utterance then play
Translator	ML Kit in APK	Once for packs	≥50 languages; AUTO swap
YAMNet / BirdNET	APK / files/models	Once for BirdNET	Duty cycle; BirdNET after YAMNet bird
Flora/fauna	APK + Coral TFLite	Once for packs	Hold frame
Horizon / G / SPD	IMU + GNSS	GNSS for speed	Uncertified; portrait native
Satellites	TLE + SGP4 in APK	Once for TLEs	Look-angles only; not radar
Stars	Bright catalog in APK	None	Major stars, not full sky
Watches	NWS / OpenSky / USGS	While Audio armed	Wi-Fi or cell
Camp mesh	Nearby Connections	Local radio	JSON alerts only

6. Installation

6.1 Sideload the APK

Allow install from the file source. Uninstall the previous Mk I build first if the signature or versionCode fights the package manager. Grant Camera, Microphone, Location, Nearby devices, and Notifications when asked.

6.2 Termux stack

Install Termux from F-Droid (not the abandoned Play listing). The APK cannot silently install Termux or pull multi-gigabyte GGUF / SD / Kokoro weights. If Termux is missing at launch, the face says so. A tricorder-start.sh helper is copied into app external files; edit model paths and run in Termux. sshd on 8022 is optional for a desk session. Status helper: ~/bin/tricorder status — expected ports 8080, 8081, 8188, 5000.

6.3 First-run earth-link fetch

If 8.8.8.8 answers, the APK fetches missing BirdNET (~25 MB), Coral flora/fauna (~12 MB), translator pair en+es (not all 50 packs — that is gigabytes), and Celestrak TLE groups. Status lines appear on the CRT. Offline, existing files stay as they are.

6.4 Permissions

Location is required for watches, GPS speed, satellite observer site, and first Nearby pairing. Disable location and SPD / sat look-angles degrade to last fix or a Newburgh default only as a last resort in the star/sat plot.

7. Guide to operations

7.1 Lights

IDLE / LISTEN / PROCESS on the bezel. CHAT, VISION, IMAGE, VOICE, TRANSLATE lamps: green idle and reachable, yellow thinking or speaking, red offline. TRANSLATE green = waiting for speech, yellow = speaking the translation, red = packing or error.

7.2 Inquiry

Speak or type. Keyboard pop is a toggle under Imagine sliders (default off). Say HELP for the spoken briefing. Say COMPUTER for the in-universe status monologue (periods stripped before Kokoro).

7.3 Sensor / Sentry / Hold

Point, Hold to freeze and classify flora/fauna. Continuous vision is screen-only. Sentry Hold after an alert is intentional.

7.4 Audio sense

Arm IV. Sliders set listen and sleep. Detections stay on the LED tape. Watches fire only in this mode. Kokoro briefings mute YAMNet until the WAV ends plus a short tail.

7.5 Universal translator

T arms the loop. SOURCE and TARGET open language wheels. AUTO green swaps direction after each line; red holds one way. Fetch pair or all packs from Backend.

7.6 Imagine sliders

Size 256 / 384 / 512. Steps typically 4–8 on SD Turbo with cfg-scale 1. Field use: one referent picture next to a translated sentence, not a print run.

7.7 Horizon, G-meter, speed

H green shows the tape in idle and Audio. Hold the unit as you read it — portrait, face to you. Nod and bank; sky/ground should move without a 90° twist. G ~ 1.00 at rest. Walk outdoors for SPD to leave zero. H red hides the tape so Audio log text has the CRT.

7.8 Satellites / Stars

S once: satellites. Check ISS, Part 97, Iridium, Globalstar (on by default). GEO and GPS are off until checked (GEO is a large file). Fetch TLEs if the plot is empty after a first boot without Wi-Fi. S twice: stars. Turn the body; names stay with the bright objects. S three times: off.

7.9 Log review, officer entry, video, play

L arms the time slider. E records a spoken or typed officer line. V starts/stops a front-camera clip. P plays the slider's current row: video opens in the system player; text and audio-log rows are spoken by Kokoro. Snaps display on the CRT.

7.10 Face sheets

Help, Credits, Warning, Manual sit under the nameplate. CLOSE is a real button. Manual is the HTML copy inside the APK (this Word document is the archival operator copy).

8. NOAA, flight, civic watch, and camp comms

8.1 Design intent

A tent unit in Audio + watches is a low-duty sentry. A fishing-hole unit receives JSON and speaks: "Remote tricorder unit detected (weather|flight|citizen) event at HH:MM. Details follow."

8.2 Enabling

Settings → Context providers. NOAA, flights, and civic/Citizen-style default ON as of 0.6.0. Arm IV. Need a network path. Camp link on at both ends.

8.3 Local and remote speech

Local: "Local tricorder detected flight event at 14:22. Details follow. UAL123 11000 m …" Remote unit uses the Remote template. Audio Sense does not talk over that sentence.

8.4 Failure modes

OpenSky rate-limits. NWS requires a User-Agent (the APK sends one). No GPS → weather/flight/civic polls skip. Waze remains a notice only.

9. Design notes

Imagine next to Translate is a generative phrasebook: mint one shared referent when the noun in speech is ambiguous. Minutes of SoC time beats a printed deck you did not pack.

Satellite work uses public TLEs and SGP4, not a phased-array. Element age of a day is normal; refresh TLEs when passes look late. Star work is a bright-star subset, not Hipparcos-complete.

The APK will not auto-install Termux or fetch Gemma/SD/Kokoro. Those weights are the operator's Termux problem on purpose: size, license, and path.

10. Competitive matrix (indicative)

Capability	Tricorder Mk I	PocketPal	Sky Map / Stellarium	Heavens-Above / Gpredict	Google Translate
On-device LLM + vision + SD + TTS	Yes (Termux)	LLM + TTS	No	No	Cloud/on-device MT only
Offline translator ≥50	ML Kit	No	No	No	Yes
YAMNet / BirdNET / flora	Yes	No	No	No	No
Horizon + G + GPS speed	Yes, uncertified	No	No	No	No
SGP4 sat plot from Celestrak	Selected groups	No	No	Yes (deeper)	No
Star map	Major stars + heading	No	Full sky	No	No
NOAA / ADS-B-class watch	NWS + OpenSky	No	No	No	No
Field book + mesh alerts	JSONL + Nearby	No	No	No	No
Certified nav / flight	No	No	No	No	No

11. Troubleshooting

Symptom	Likely cause	Action
All lamps red, no button works	JS parse break in an old build	Install 0.6.1 or later
CLOSE on Help does nothing	Pre-0.6.2 sheets after script	Install 0.6.2+
Horizon needs 90° twist	Landscape-era mapping	0.6.2+ portrait remap
SPD stays —	No GNSS fix	Outdoors, location on
Sat plot empty	No TLE files	Wi-Fi init or Fetch TLEs
Watch talks then "Speech detected"	Mic heard Kokoro	0.6.5 hold; use 0.6.2+
Translator beeps after Inquiry	Recognizer busy	Fetch pair again; 0.5.x backoff
Imagine 400	sd-server JSON shape	Keep OpenAI images body; sliders 4–8 steps
Termux alert at boot	F-Droid Termux missing	Install Termux, start the four servers
Camp silent	One unit Camp off	Both on, location + Nearby granted

12. Maintenance

Refresh TLE packs when earth link is cheap; elements rot. Re-fetch BirdNET/flora only if files were wiped. Do not treat field.jsonl as a durable archive — export off the device when the survey ends. Uninstall between major signature changes.

13. Document control

Rev	Date	Notes
0.4.1	2 Sep 2026	Field book, mesh notes
0.5.4	4 Sep 2026	Translator, horizon, NOAA/OpenSky, FAA warning
0.6.4	5 Sep 2026	G-meter, GPS speed, Sat/Stars, civic watch, first-run fetch, play-log, Help/Credits/Warning

Prepared for operator use. N2NHU Labs, Newburgh, New York 12550.